Complex Numbers

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

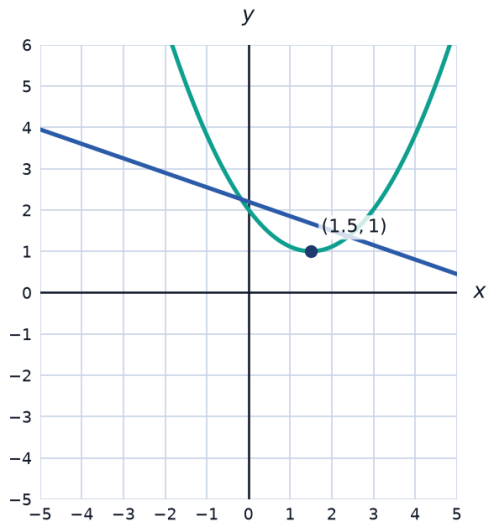
1 **[Foundational]** • **Target: Powers of i**

Evaluate each power of i, using $i^2 = -1$ and the fact that powers of i repeat every 4:

a i^3 **b** i^4 **c** i^7 **d** i^{22}

2 **[Foundational]** • **Target: Multiplication**

Multiply $(5 - 2i)(5 + 2i)$, and explain why the result has no imaginary part.



3 **[Foundational]** • **Target: Conjugate and modulus**

Show that $z + \bar{z}$ is always real for $z = a + bi$.

4 **[Foundational]** • **Target: Division**

Simplify $\frac{3 + 4i}{1 - 2i}$ by multiplying top and bottom by the conjugate of the denominator.

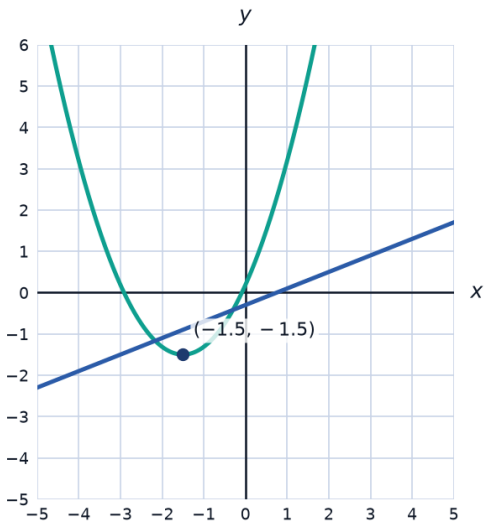
5 **[Foundational]** • **Target: Quadratic equations with complex roots**

Solve $x^2 - 6x + 13 = 0$ using the quadratic formula.

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6 [Foundational] • Target: Quadratic equations with complex roots

Solve the quadratic equation $2x^2 - 6x + 9 = 0$ over the complex numbers, and state the sum and product of the roots.



7 [Foundational] • Target: Powers of i

The polynomial $P(x) = 2x^3 + kx^2 - 7x + 6$ has $(x - 2)$ as a factor. (a) Find the value of k. (b) Factor $P(x)$ completely over the real numbers and list all its roots.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Addition and subtraction

Express $z = \frac{4+7i}{1-2i}$ in standard form $a + bi$, and compute its modulus squared $|z|^2$.

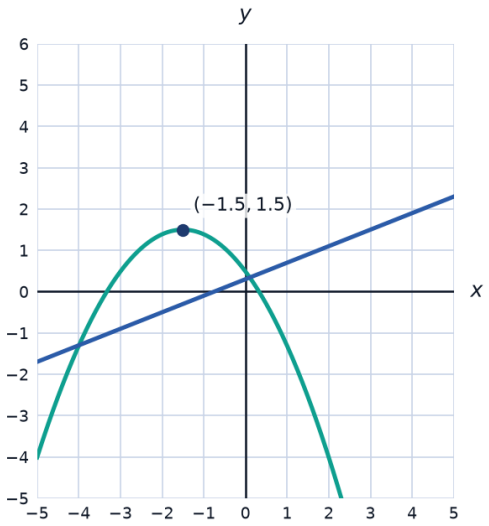
9 [Intermediate] • Target: Multiplication

Find the Cartesian equation in terms of x and y of the locus of complex numbers $z = x + yi$ satisfying the equation $|z - 3| = |z + 3i|$, and describe its geometric shape.

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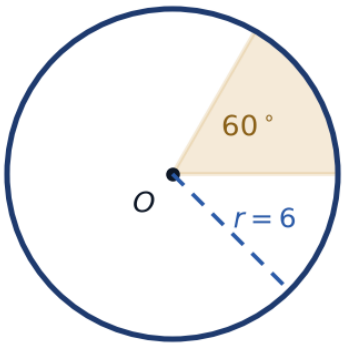
10 **[Intermediate]** • **Target: Conjugate and modulus**

Compute $(1 - i)^{10}$ in exact rectangular form $a + bi$ using polar representation.



11 **[Intermediate]** • **Target: Division**

Find the modulus and principal argument of $z = -2\sqrt{3} + 2i$, and express z in polar form $r(\cos\theta + i\sin\theta)$.



12 **[Intermediate]** • **Target: Quadratic equations with complex roots**

Consider the function $f(x) = \sqrt{2x - 5} + 3$. (a) Find the domain and range of $f(x)$. (b) Find the explicit equation for the inverse function $f^{-1}(x)$, including its restricted domain.

13 **[Intermediate]** • **Target: Powers of i**

Solve the logarithmic equation for x: $\log_3(x + 2) + \log_3(x - 4) = 3$

14 **[Intermediate]** • **Target: Addition and subtraction**

Solve the rational equation for all real values of x: $\frac{x}{x-2} + \frac{1}{x-4} = \frac{2}{(x-2)(x-4)}$

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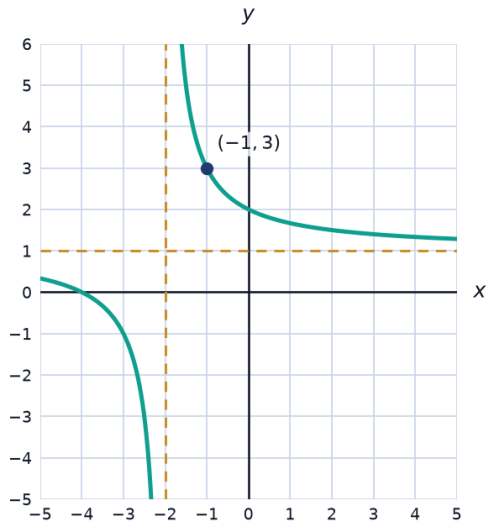
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Multiplication** • **Cross-Lesson: 6.01 Limits**

Calculate the exact value of $\cos(75^\circ)$ using angle addition, and evaluate $\tan(2\theta)$ given that $\tan\theta = \frac{2}{3}$.

16 **[Challenging]** • **Target: Conjugate and modulus** • **Cross-Lesson: 2.01 Matrices & Vectors**

Evaluate the limit using algebraic conjugate rationalization: $\lim_{x \to 4} \frac{\sqrt{x} - 2}{x - 4}$



17 **[Challenging]** • **Target: Division** • **Cross-Lesson: 3.01 Sequences & Series**

Find all complex solutions to the cubic equation $z^3 + 8 = 0$.

18 **[Challenging]** • **Target: Quadratic equations with complex roots** • **Cross-Lesson: 4.01 Advanced Functions & Equations**

Identify the geometric shape in the complex plane described by the equation $|z - 4| + |z + 4| = 10$, and calculate the length of its minor axis.

19 **[Challenging]** • **Target: Powers of i** • **Cross-Lesson: 5.01 Trigonometry & Polar Coordinates**

Simplify the complex quotient into standard form $a + bi$: $\frac{3 + 4i}{1 + 2i}$

20 **[Challenging]** • **Target: Addition and subtraction** • **Cross-Lesson: 6.01 Limits**

If r_1 and r_2 are the roots of $3x^2 - 8x + 5 = 0$, find the exact value of $\frac{1}{r_1} + \frac{1}{r_2}$ without solving for the roots individually.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Powers of i

Evaluate i^{130}, using the fact that powers of i repeat every 4.

a. i

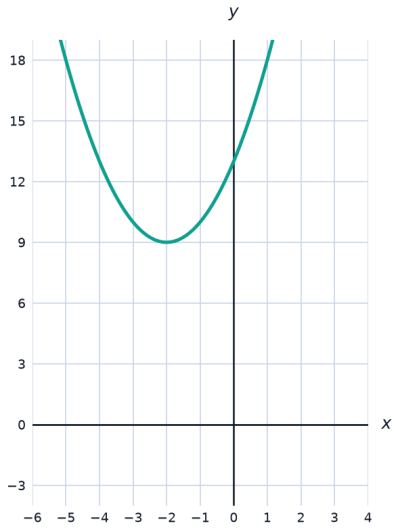
b. $-i$

c. 1

d. -1

22 [Foundational] • Target: Addition and subtraction

Find the product of the roots of $x^2 + 4x + 13 = 0$.



a. 13

b. 15

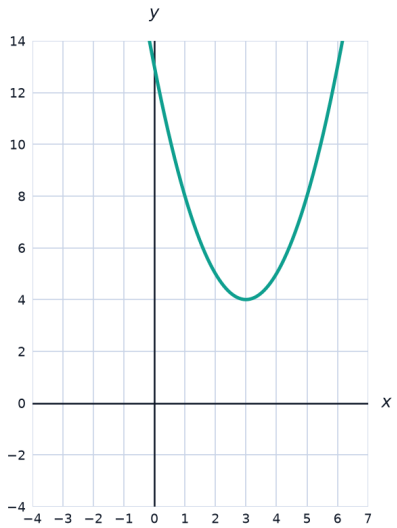
c. 4

d. -13

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23 [Foundational] • Target: Multiplication

Find the sum of the two roots of $x^2 - 6x + 13 = 0$.



a. 6

b. 3

c. 13

d. $4i$

24 [Foundational] • Target: Conjugate and modulus

What is the complex conjugate of $5 - 2i$?

a. $5 + 2i$

b. $-5 + 2i$

c. $5 - 2i$

d. $-5 - 2i$

25 [Foundational] • Target: Division

For which value of k does $x^2 + kx + 9 = 0$ have complex (non-real) roots?

a. $k = 5$

b. $k = 7$

c. $k = 6$

d. $k = -8$

26 [Foundational] • Target: Quadratic equations with complex roots

Find the complex conjugate of $7 - 3i$.

a. 7+3i

b. -7-3i

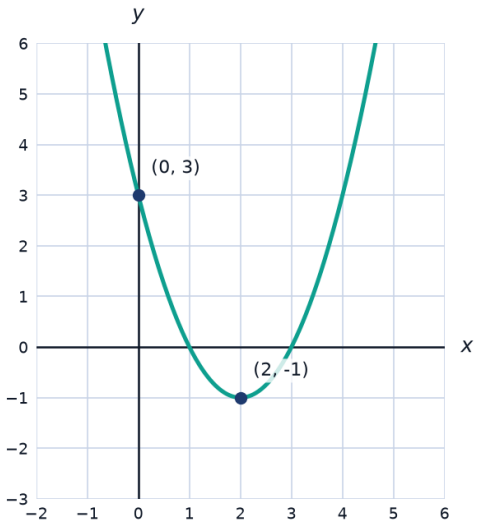
c. -7+3i

d. 7-3i

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27 [Foundational] • Target: Powers of i

The graph of the quadratic function $f(x) = ax^2 + bx + c$ is shown in the standard (x, y) coordinate plane below, with vertex $(2, -1)$ and y-intercept $(0, 3)$. The function g is defined by $g(x) = f(x + k) - 8$, where k is a constant. If the graph of g passes through the origin $(0, 0)$, which of the following could be the value of k?



a. -5

b. -3

c. 1

d. 5

28 [Foundational] • Target: Addition and subtraction

Let z be a complex number satisfying the equation $|z - 12| + |z + 12| = 26$. What is the maximum possible value of $|z - 5i|$?

a. $\frac{169}{12}$

b. $\sqrt{194}$

c. 18

d. 10

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 **[Intermediate]** • **Target: Multiplication**

In the complex plane, let z be a complex number that satisfies the equation $|z-4|+|z+4|=10$. What is the maximum possible value of $|z-3i|$?

a. $\frac{25}{4}$

b. 6

c. $\sqrt{34}$

d. $\frac{13}{2}$

30 **[Intermediate]** • **Target: Conjugate and modulus**

What is the minimum value of the function $f(x)=\sqrt{x^2-4x+13}+\sqrt{x^2-14x+130}$ for all real numbers x?

a. 12

b. $4\sqrt{10}$

c. 13

d. $\sqrt{170}$

31 **[Intermediate]** • **Target: Division**

For what real value of k is the complex expression $\frac{k+8i}{3-i}+\frac{2-ki}{3+i}$ a real number?

a. 11

b. 1

c. 6

d. -11

32 **[Intermediate]** • **Target: Quadratic equations with complex roots**

In the complex plane, the complex numbers z and w satisfy $|z|=5$, $|w|=7$, and $|z+w|=10$. What is the value of $|z-w|$?

a. $4\sqrt{3}$

b. $2\sqrt{6}$

c. $4\sqrt{6}$

d. $2\sqrt{37}$

33 **[Intermediate]** • **Target: Powers of i**

A function f is defined for all real numbers except $x=0$ and $x=1$. For all values of x in its domain, f satisfies the functional equation $f(x)+f\left(\frac{x-1}{x}\right)=1+x$. What is the value of $f(2)$?

a. $\frac{3}{4}$

b. $\frac{5}{4}$

c. $\frac{7}{4}$

d. $\frac{9}{4}$

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34 [Intermediate] • Target: Addition and subtraction

In the complex number system, let $i^2 = -1$. For what real value of k is the expression $\frac{6+ki}{3+2i}$ a purely imaginary number?

a. -9

b. -4

c. 4

d. 9

35 [Intermediate] • Target: Multiplication

For the imaginary unit i, where $i^2 = -1$, the expression $\frac{k+6i}{3-2i}$ is equal to a real number for some real constant k. What is the value of k?

a. -9

b. -4

c. 4

d. 9

36 [Intermediate] • Target: Conjugate and modulus

For the imaginary unit i, where $i^2 = -1$, which of the following is equivalent to the expression $(5 - 2i)(3 + 4i)$?

a. $23 + 14i$

b. $7 + 14i$

c. $23 + 26i$

d. $7 - 14i$

37 [Intermediate] • Target: Division

For the complex number $i = \sqrt{-1}$, which of the following is equivalent to the expression $(5 - 3i) - (2 - 7i)$?

a. $3 + 4i$

b. $3 - 10i$

c. $7 + 4i$

d. $3 - 4i$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Quadratic equations with complex roots • Cross-Lesson: 4.01 Advanced Functions & Equations

The function f is defined by $f(x) = \frac{3x+2}{x-4}$ for all real numbers $xeq4$. What is the value of $f^{-1}(2)$?

a. -10

b. -6

c. -4

d. 6

39 [Challenging] • Target: Powers of i • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

For the imaginary unit i such that $i^2 = -1$, what is the product $(4 - 3i)(2 + 5i)$?

a. $23 + 14i$

b. $-7 + 14i$

c. $23 - 14i$

d. $8 - 15i$

40 [Challenging] • Target: Addition and subtraction • Cross-Lesson: 6.01 Limits

For the imaginary unit i, where $i^2 = -1$, which of the following complex numbers is equivalent to $(4 + 3i)(2 - 5i)$?

a. $23 - 14i$

b. $23 + 14i$

c. $-7 - 14i$

d. $-7 + 14i$

41 [Challenging] • Target: Multiplication • Cross-Lesson: 2.01 Matrices & Vectors

Find all real roots of $x^3 + 8x^2 + 19x + 12 = 0$.

a. x=-4,\ -3,\ -1

b. x=-4,\ -3,\ 0

c. x=-4

d. x=-4,\ -3

42 [Challenging] • Target: Conjugate and modulus • Cross-Lesson: 3.01 Sequences & Series

Solve $x^2 + 2x + 5 = 0$. Which root has positive imaginary part?

a. $-1 + 2i$

b. $1 + 2i$

c. $-1 - 2i$

d. $2 + i$

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43 [Challenging] • Target: Division • Cross-Lesson: 4.01 Advanced Functions & Equations

Is $f(x) = |x|$ one-to-one over all real numbers?

a. No

b. Yes

c. Cannot be determined

d. Only for $x < 0$

44 [Challenging] • Target: Quadratic equations with complex roots • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

What is the imaginary part of $z = 6 - 7i$?

a. -7

b. 6

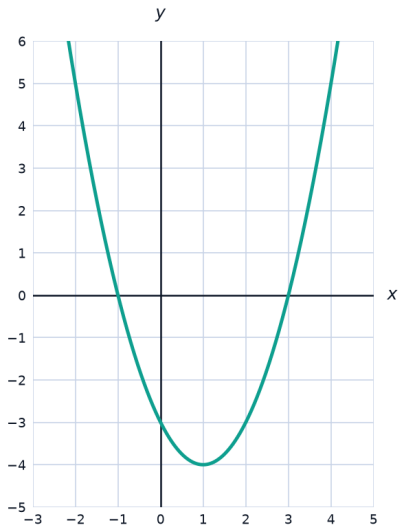
c. 7

d. -6

45 [Challenging] • Target: Powers of i • Cross-Lesson: 6.01 Limits

The graph of the function $f(x) = (x - 1)^2 - 4$ is shown in the standard (x, y) coordinate plane below.

What is the range of f?



a. $[-4, \infty)$

b. $[1, \infty)$

c. $(-\infty, -4]$

d. $(-\infty, \infty)$

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